

Example 1 : Basic Contradiction

$$\frac{\begin{array}{c} \frac{\frac{\frac{\frac{}{\neg p \wedge (q \wedge \neg q)}{\neg q}}{q \wedge \neg q} [\wedge\text{-elim-2}]}{\neg q} [\wedge\text{-elim-1}]}{q} [\wedge\text{-elim-2}] \\ \neg q \\ \hline false \\ [contradiction] \\ \hline p \\ [false\text{-elim}] \end{array}}{\neg p \wedge (q \wedge \neg q) \Rightarrow p} [\Rightarrow\text{-intro}^{[1]}]$$
$$\frac{\frac{\frac{\frac{p \wedge \neg p^{\neg[1]}}{p \wedge \neg \text{elim-1}}}{\neg p}}{\text{contradiction}}}{\text{false-elim}}}{q} \frac{}{(p \wedge \neg p) \Rightarrow q} [\Rightarrow \text{-intro}^{[1]}]$$
$$\frac{\frac{\lceil p \wedge \neg p \rceil^{[1]}}{false} \text{ [contradiction]}}{p \wedge \neg p \Rightarrow false} [\Rightarrow \text{-intro}^{[1]}]$$
$$\frac{\frac{\frac{\overline{false}}{\neg p} \quad [\neg\text{-intro}^{[2]}]}{\neg(\neg p)} \quad [\Rightarrow\text{-elim}]}{\overline{\neg p \Rightarrow false} \Rightarrow \neg(\neg p)} \quad [\Rightarrow\text{-intro}^{[1]}]$$

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Example 5 : Modus Tollens via Contradiction

Prove: $(p \Rightarrow q) \wedge \neg q \Rightarrow \neg p$

$$\frac{\frac{\neg q \quad [\Rightarrow\text{-elim}]}{q} \quad \frac{\text{contradiction}}{\text{false}} \quad [\text{assumption}]}{\frac{p}{\neg p} \quad [\neg\text{-intro}^{[2]}]} \quad [\Rightarrow\text{-intro}^{[1]}]$$

Classic modus tollens pattern using contradiction.

Example 6 : Disjunction from Contradiction

Prove: $\neg p \Rightarrow (p \Rightarrow q)$

$$\frac{\frac{\frac{\neg p \quad \neg^{[2]}}{\text{false}} \quad [\text{contradiction}]}{q} \quad [\text{false-elim}]}{(p \Rightarrow q) \quad [\Rightarrow\text{-intro}^{[2]}]} \quad [\Rightarrow\text{-intro}^{[1]}]$$

From $\neg p$, we can prove $p \Rightarrow q$ for any q.

Example 7 : Double Negation Elimination

Prove: $\neg(\neg p) \Rightarrow p$ (requires classical logic)

$$\frac{\frac{\frac{\neg \neg p \quad \neg^{[2]}}{\text{false}} \quad [\text{contradiction}]}{p} \quad [\text{false-elim}]}{\neg(\neg(\neg p)) \quad [\neg\text{-intro}^{[2]}]} \quad [\Rightarrow\text{-intro}^{[1]}]$$

In classical logic, not $\neg p$ implies p. This requires excluded middle or equivalent axiom.

Example 8 : Reductio ad Absurdum

Prove: $(\neg p \Rightarrow (q \wedge \neg q)) \Rightarrow p$

$$\frac{\frac{\frac{\frac{\neg \neg p \quad \neg^{[2]}}{q \wedge \neg q} \quad [\Rightarrow\text{-elim}]}{q \wedge \neg q} \quad [\wedge\text{-elim-1}]}{\frac{q}{\neg q} \quad [\wedge\text{-elim-2}]} \quad [\text{contradiction}]}{\frac{\text{false}}{p} \quad [\text{false-elim}]} \quad [\neg\text{-intro}^{[2]}]$$

$$\frac{\neg(\neg p)}{(\neg p \Rightarrow (q \wedge \neg q)) \Rightarrow \neg(\neg p)} \quad [\Rightarrow\text{-intro}^{[1]}]$$

Reductio ad absurdum: if assuming not p leads to absurdity, then p holds.

Example 9 : Contradiction with Universal Quantifier

Prove: $(\forall x \bullet p(x)) \wedge (\exists x \bullet \neg ((p(x))))$ is contradictory.

$$\frac{\frac{\frac{\frac{\frac{\frac{\frac{\Gamma(\forall x \bullet p(x)) \wedge (\exists x \bullet \neg((p(x))))}{\forall x \bullet p(x)} [\wedge\text{-elim-1}]}{\exists x \bullet \neg((p(x)))} [\wedge\text{-elim-2}]}{\neg((p(a)))} [\exists\text{ elim, fresh } a]}{\frac{p(a)}{false}} [\forall\text{-elim}]}[\text{contradiction}]}}{\frac{\Gamma(\forall x \bullet p(x)) \wedge (\exists x \bullet \neg((p(x))))}{((\forall x \bullet p(x)) \wedge (\exists x \bullet \neg((p(x)))))) \Rightarrow false} [\Rightarrow\text{-intro}^{[1]}]}$$

If something holds for all x , it cannot fail for some x .

Example 10 : Proving Uniqueness by Contradiction

Prove: if f is injective, then $f(x) = f(y) \Rightarrow x = y$.

$$\frac{\frac{\frac{\top x \neq y \top^{[2]}}{f(x) \neq f(y)} \text{ [injective property]}}{\text{false}} \text{ [contradiction with } f(x) = f(y)]}{x = y} \text{ [}\neg\text{-intro}^{[2]}]$$

$$\frac{}{(injective(f) \wedge f(x) = f(y)) \Rightarrow x = y} \text{ [}\Rightarrow\text{-intro}^{[1]}]$$

Uses contradiction to prove equality.

Example 11 : Case Analysis Leading to Contradiction

Prove: $p \vee q, \neg p, \neg q \Rightarrow false$

$$\frac{\frac{\overline{false} \text{ [contradiction with } \neg p]}{\overline{false}} \quad \frac{\overline{false} \text{ [contradiction with } \neg q]}{\overline{false} \text{ [\ve elim from } p \vee q]}}{\overline{false} \text{ [\Rightarrow -intro}^{[1]}]}} \frac{}{((p \vee q) \wedge \neg p \wedge \neg q) \Rightarrow false}$$

Both cases lead to contradiction, so the premises are inconsistent.

Example 12 : Best Practices for Contradiction Proofs

When using proof by contradiction: 1. Clearly mark the assumption you're contradicting with [assumption N] 2. Show explicitly where false is derived 3. Use [not intro from N] to discharge the assumption 4. Document the contradiction (what conflicts with what) 5. In natural deduction, false elim lets you conclude anything 6. Remember: contradiction is a classical technique (not constructive)